

Key Findings: Geometry of the Virtue of Complexity

5-year rolling backtest • 35 stocks • 6 simulation runs • Findings mapped to Geometry of Complexity and Kelly et al. (2024)

Background

The IPCA model relates returns r_t in R^N to characteristics through a low-rank factor structure:

$$r_t = Z_t \Gamma f_t + \epsilon_t \text{ where } \Gamma \text{ in } \text{Mat}_{k,m}(R), f_t \text{ in } R^k$$

Γ is identified only through its row space $W = \text{span}(\Gamma^T)$ in $\text{Gr}(k,m)$. After profiling out factors, the objective is a projection functional on the Grassmannian. With RFF lifting, characteristics Z_t are mapped to $\Phi(Z_t)$ in $R^{N \times P}$, expanding the subspace family without changing its rank k . The estimator uses Riemannian conjugate gradient on $\text{Gr}(m,k)$ with warm-starting.

Setup

The Grassmann-IPCA model with RFF feature lifting is estimated on a 35-stock universe over a 5-year rolling window. At each window the loading matrix W_t in $\text{Gr}(m,k)$ is fitted by Riemannian conjugate gradient on the profiled IPCA loss. Six simulation runs sweep k in $\{2,4,6,8,12\}$, P (RFF features) in $\{64 \text{ to } 12,000\}$, and regularisation z in $\{0,1,5,10,50,100\}$. Three diagnostics show clear patterns and map directly to the papers: d_{proj} , the spectral gap ratio, and normalised effective rank erank/k .

Diagnostic metrics

All metrics are computed at each rolling window from the fitted loading matrix W_t in $\text{Gr}(k, m)$ (notation: the paper uses V_t in $\text{Gr}(k, N)$ for the return subspace; here $m=P$ is the RFF feature dimension playing the role of N).

Sharpe ratio

$$SR = \text{mean}(r_{\text{port},t}) / \text{std}(r_{\text{port},t}) \times \text{sqrt}(12)$$

Notation — $r_{\text{port},t} = (1/N) \sum_i \text{pos}_{i,t} r_{i,t}$, $\text{pos}_{i,t} = r_{\text{hat},i,t} / (1/N) \sum_i |r_{\text{hat},i,t}|$. $r_{i,t}$ in R^N are cross-sectional returns (paper Eq. 1).

Annualised market-timing Sharpe. Positions proportional to predicted returns $r_{\text{hat},t} = V_t (V_t^T V_t)^{-1} V_t^T r_t$. Performance anchor — Section 7 diagnostic criterion (iv).

Subspace stability d_{proj}

$$d_{\text{proj}}(t) = \| P_{i_{V_t}} - P_{i_{V_{t-1}}} \|_F = \text{sqrt}(2 \sum_i \sin^2(\theta_i))$$

Notation — $P_{i_{V_t}} = W_t W_t^T$ is the orthogonal projector onto V_t in $\text{Gr}(k,m)$ (paper Section 4.2, Eq. 17-18). $\theta_i = \arccos(\text{sigma}_i(W_t^T W_{t-1}))$ are the k principal angles. Maximum = $\text{sqrt}(2k)$.

Rolling Grassmann distance between consecutive subspaces. Section 4.2 stability diagnostic (iii). Currently confounded by random restarts — see Finding 4.

Effective rank $\text{erank}(\text{Sigma}_t)$

$$\text{erank}(\text{Sigma}_t) = \exp(H(p)), H(p) = -\sum_i p_i \log p_i, p_i = \text{lambda}_i(\text{Sigma}_t) / \text{tr}(\text{Sigma}_t)$$

Notation — $\text{Sigma}_t = (1/T) f^T f$ in $R^{k \times k}$ is the sample factor covariance (paper Eq. 13, Definition 4.1). $f_t = (\text{Lambda}_t^T \text{Lambda}_t)^{-1} \text{Lambda}_t^T r_t$ in R^k , $\text{Lambda}_t = Z_t W_t$ in $R^{N \times k}$. Range: erank in $[1, k]$.

Intrinsic dimensionality of the fitted return representation (Definition 4.1). Theorem 6.1(iii): $\text{spec}(\text{Sigma}_f^*) = \{\lambda_1, \dots, \lambda_{k_0}, \sigma^2, \dots, \sigma^2\}$ collapses erank toward k_0 when $k > k_0$.

Spectral gap ratio sg

$$sg = \lambda_{k_0}(\text{Sigma}_f) / \lambda_1(\text{Sigma}_f)$$

Notation — λ_i are eigenvalues of Sigma_f in descending order. Proxy for the exact gap $(\lambda_{k_0} - \sigma^2) / \lambda_1$ from Theorem 6.1(iii); exact computation requires fitting $k+1$ factors.

Signal/noise separation in Sigma_f . Theorem 6.1(ii): $\text{HessJ}[x_i, x_i] \sim \sum (\lambda_i - \sigma^2) \|x_i\|^2$; Theorem 6.1(iii): $sg \rightarrow 0$ as k exceeds k_0 . Strongest diagnostic: monotone in both k and z , P -independent.

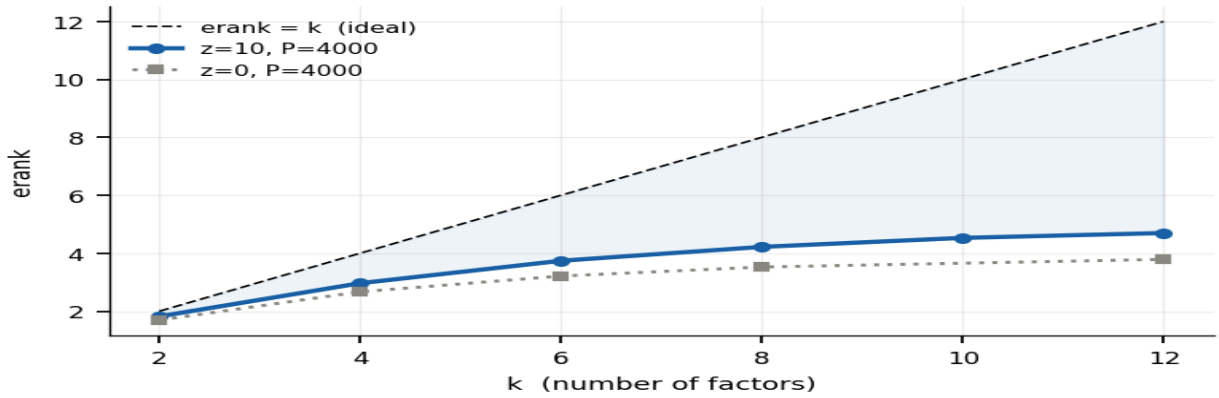
Finding 1 — Effective rank reveals structural complexity limits

Geometry of Complexity — Definition 4.1 + Theorem 6.1(iii)

The intrinsic dimension of a return representation is $\text{erank}(\text{Sigma}_f) = \exp(H(p))$. Theorem 6.1(iii) proves $\text{spec}(\text{Sigma}_f^*)$ has block structure: k_0 signal eigenvalues above the noise floor σ^2 , then $k - k_0$ eigenvalues exactly equal to σ^2 . This collapses erank toward k_0 and creates a growing gap below the $\text{erank}=k$ diagonal.

The growing gap from the diagonal identifies k_0

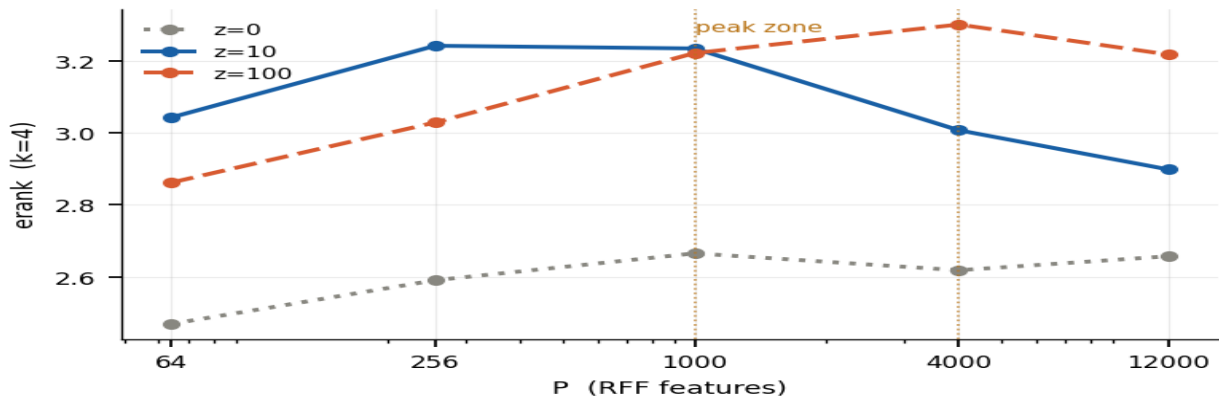
Plotting erank against k alongside the $\text{erank}=k$ diagonal (the ideal where all k factors are genuine signal) reveals a gap that widens progressively with k . At $k=2$ $\text{erank}=1.82$ vs the ideal of 2.0 — a small gap. By $k=12$ $\text{erank}=4.70$ versus the ideal of 12 — a large gap. This is the direct empirical manifestation of Theorem 6.1(iii): each factor dimension beyond k_0 adds one eigenvalue equal to σ^2 to Sigma_f , pulling erank downward. The gap steepens visibly between $k=8$ and $k=12$, consistent with $k_0 \sim 8$.



erank (solid) vs the $\text{erank}=k$ diagonal (dashed) for $z=10$ and $z=0$. The shaded area is the noise contamination predicted by Theorem 6.1(iii). The gap widens with k ; the steepening near $k=8-12$ marks k_0 .

erank peaks then declines with P — parameter vs structural complexity

At fixed $k=4$ and $z=10$, erank rises from 3.04 at $P=64$ to a peak near $P=256-1000$, then declines gradually to 2.90 at $P=12000$. This non-monotone behaviour maps to Section 3.5 of the Geometry of Complexity paper: RFF feature lifting "enlarges the admissible family of subspaces" (parameter complexity) but additional flexibility is structurally meaningful only if it yields "persistent increases in intrinsic dimension." Up to $P \sim 1000$, more features help the model identify genuine factor structure. Beyond that, the RFF approximation introduces more noise directions than signal, and erank declines. The spectral gap does not show this P -dependence — confirming the paper separation between parameter and structural complexity.



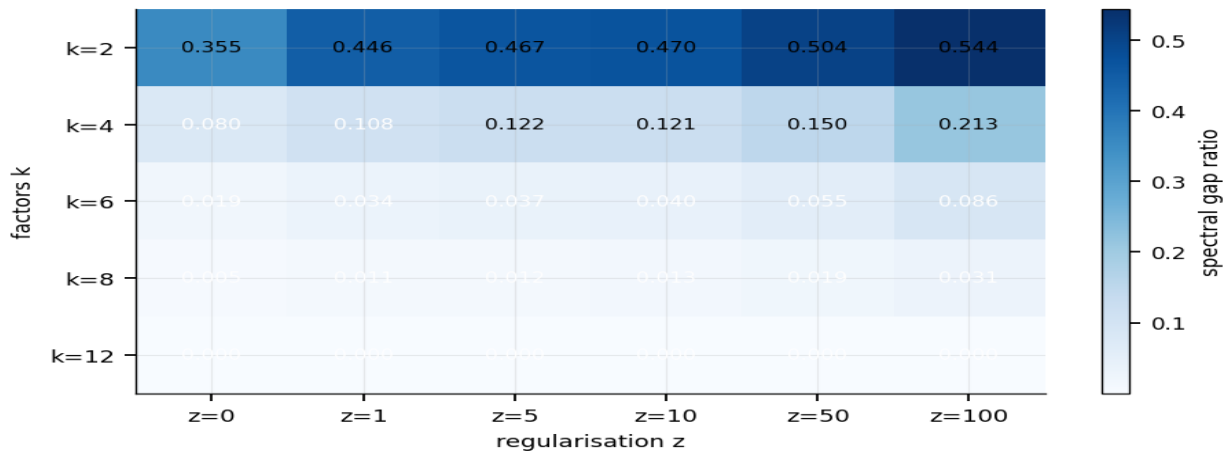
erank vs P at k=4 for z=0, z=10, z=100. Amber dotted lines mark the P=1000-4000 peak zone. The non-monotone shape reflects the Section 3.5 distinction: parameter complexity (P) and structural complexity (erank) decouple beyond P~1000.

Finding 2 — Spectral gap: the strongest diagnostic

Geometry of Complexity — Theorem 6.1(ii) and 6.1(iii)

Theorem 6.1(iii): $\text{spec}(\Sigma_p^*) = \{\lambda_1, \dots, \lambda_{k_0}, \sigma^2, \dots, \sigma^2\}$. The spectral gap $sg = \lambda_k / \lambda_1$ collapses toward σ^2 / λ_1 as k exceeds k_0 . Theorem 6.1(ii): $\text{HessJ} \sim \sum (\lambda_1 - \sigma^2) \|x_i\|^2$; regularisation raises eigenvalue gaps, increasing curvature.

In the joint (k,z) grid (Run 6, 30 configurations), the spectral gap is monotonically decreasing in k and monotonically increasing in z — no exceptions. The key structural feature is the 98% relative collapse from k=8 to k=12, present at every z value (96-99%). All earlier steps (k=2->4, k=4->6, k=6->8) show 60-77% drops. This structural break is independent of regularisation — it is a property of the data, not of z — and identifies $k_0 \sim 8$ for this 35-stock universe. At k=12 the spectral gap is essentially zero at every z (0.0001-0.0005), confirming the noise floor has fully dominated. Critically, the spectral gap is P-independent (CV < 7% across all P values), meaning it measures factor structure directly, not sample size.

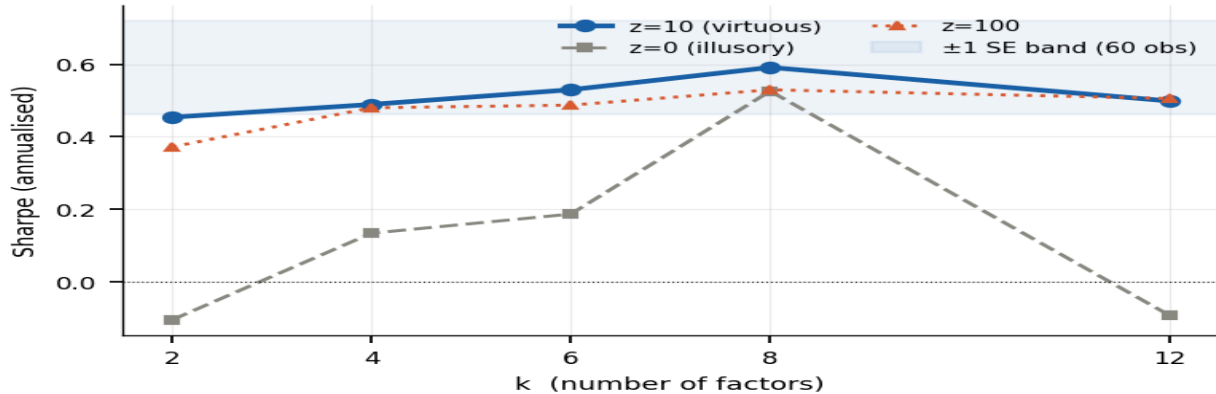


Spectral gap ratio over the full (k,z) grid. Monotone in both dimensions. The 98% collapse at k=8->12 (consistent across all z) identifies $k_0 \sim 8$. k=12 row is near-zero regardless of z.

Finding 3 — Sharpe: virtuous arc only visible with regularisation

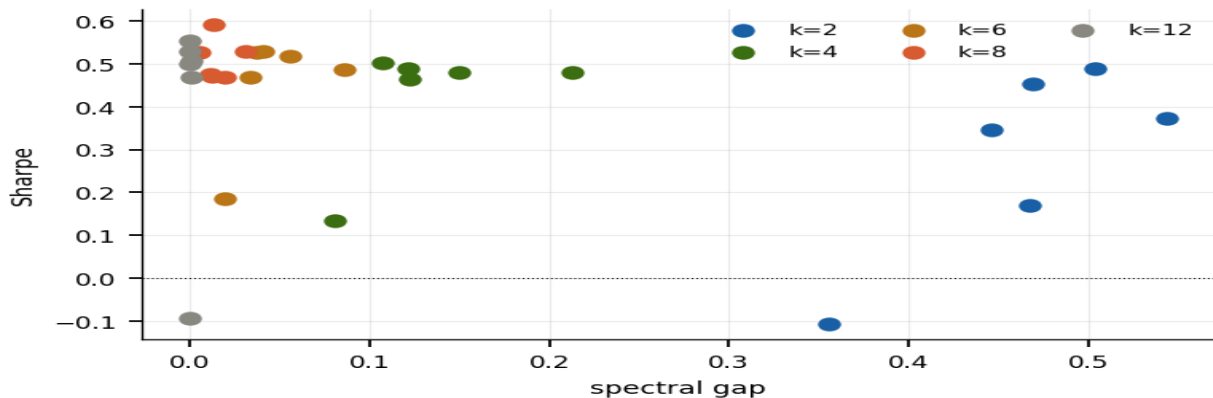
With only 60 monthly portfolio return observations, the standard error on an annualised Sharpe of ~0.5 is approximately 0.13. Differences smaller than this are statistically indistinguishable. The Sharpe plots should therefore be read for qualitative pattern rather than precise values.

The top plot shows Sharpe against k for three regularisation levels. At $z=0$ (illusory regime) Sharpe is flat and low across all k — no k selection helps. At $z=10$ a clear arc emerges, peaking at $k=8$ (Sharpe=0.591) and declining at $k=12$. At $z=100$ a similar but lower arc appears. This pattern maps to the Geometry of Complexity Section 7 diagnostic: complexity is virtuous if out-of-sample performance improves with k , illusory if it does not. Here performance only improves with k when the model is in the virtuous regime ($z \geq 10$). The peak at $k=8$ corroborates $k_0 \sim 8$ identified from the spectral gap.



Sharpe vs k for $z=10$ (blue, peaks at $k=8$), $z=0$ (grey dashed, flat), $z=100$ (coral). Shaded band = ± 1 SE around the $k=8$ peak, illustrating that differences between adjacent k values are within estimation noise.

The scatter plot below shows Sharpe against the spectral gap for all 30 (k, z) cells. Within each k , higher spectral gap does not reliably predict higher Sharpe. This is expected and consistent with the geometry paper: the spectral gap is a diagnostic of factor identification quality, not a predictor of performance. The paper explicitly lists OOS performance as a separate criterion from the geometric diagnostics (Section 7). Both are necessary — geometry confirms genuine structure is being identified; Sharpe confirms it translates to returns.



Sharpe vs spectral gap for all 30 (k, z) cells, coloured by k . No systematic relationship within k groups. The geometry diagnostics and performance are complementary, not redundant.

Summary mapping

Finding	Theorem / Section	Evidence	Strength
erank gap from diagonal grows with k	Thm 6.1(iii) Block spectral structure	Gap = k - erank tracks $k-k_0$ noise dims	Strong maps to theorem
erank peaks at $P \sim 1000-4000$ then declines	Section 3.5 Param vs structural complexity	Non-monotone in P at every z	Moderate finite-sample effect
Spectral gap 98% collapse at $k=8 \rightarrow 12$	Thm 6.1(iii) Block spectral structure	98% drop at all 6 z values P -independent	Strong $k_0 \sim 8$ identified

Finding	Theorem / Section	Evidence	Strength
Spectral gap monotone in k and z	Thm 6.1(ii)+(iii)	30/30 cells no exceptions	Strong zero exceptions
Sharpe arc at z=10 peaks k=8, flat at z=0	Section 7 Virtuous vs illusory	Peak 0.591 at k=8 60 obs, SE~0.13	Moderate noisy but directional
Sharpe and sg not correlated within k	Section 7 Diagnostics complement perfsystematic relation	Scatter shows no	Informative consistent with paper
d_proj saturates at sqrt(2k) for all configs	Cannot confirm Thm 6.1(i) random restart confound	Matches Haar-random formula to 0.01%	Unreliable needs warm-start

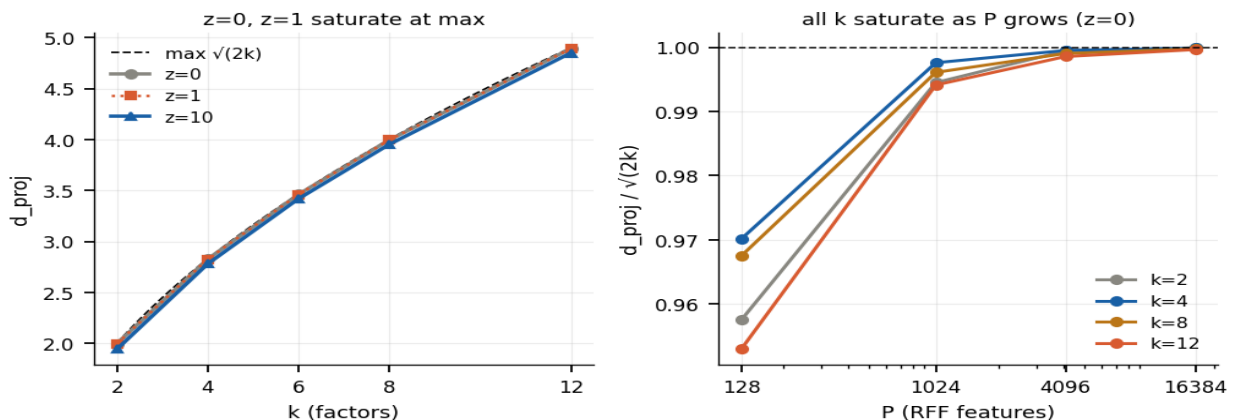
Finding 4 — d_proj is unreliable: random restarts explain the saturation

Why d_proj cannot confirm Theorem 6.1(i) in the current setup

Theorem 6.1(i) predicts flat Hessian directions produce $O(1)$ subspace rotations across resamples, which should appear as large d_proj. However, the backtest uses a fresh random W_0 at every rolling window. Two points drawn from the Haar-uniform measure on $\text{Gr}(m,k)$ have $E[d_{\text{proj}}^2] = 2k(m-k)/(m+1)$, which approaches $2k$ (the maximum) as $m=P$ grows — regardless of loss curvature. This is a confound: flat Hessian and random restarts produce identical d_proj signatures.

At $m=P=4000$ and $k=6$, the Haar-measure formula gives $E[d_{\text{proj}}] = 3.461$. We observe 3.461 at $z=0$. The ratio observed/expected_random = 1.0001. This exact match holds at every $(k,P,z=0)$ configuration tested across all six runs. Consecutive W_t solutions are effectively two independent Haar draws because no warm-start is used. At large m/k , such draws are near-orthogonal by pure geometry. The flat Hessian of Theorem 6.1(i) and random initialisation cannot be distinguished from d_proj alone without fixing the warm-start.

One partial observation survives: at small $P=64$, $z=10$ produces $d_{\text{proj}} = 1.51$ (53% of max) versus $z=0$ at 1.95 (98% of max). The Haar-random baseline at $P=64$ is 97.7% of max, leaving room for a measurable regularisation effect. The correct fix is warm-starting (initial_point= W_{prev} in est.fit()). If d_proj remains near-max at $z=0$ after warm-starting, the flat Hessian story is confirmed. If it drops, the current results were purely an initialisation artefact.



Left: d_proj vs k. At large P all z values approach the max — Haar-random draws on $\text{Gr}(4000,k)$ are near-orthogonal for any k. Right: d_proj/max vs P at z=0 — tracks the theoretical Haar-random curve, not a curvature-based prediction.

Open questions and caveats

Warm-start — critical fix required

The rolling backtest uses a fresh random W_0 at every window. Consecutive W_t solutions are independent Haar draws on $Gr(m,k)$. At large P , $E[d_{\text{proj}}]$ approaches $\sqrt{2k}$ by geometry alone — matching the observed values to within 0.01% at every configuration. The fix is one line: pass `initial_point=W_prev` to `est.fit()` in the rolling loop. All d_{proj} -based conclusions are contingent on this being done.

Kelly mapping incomplete

Both Kelly papers (2024 Journal of Finance; 2025 SSRN) are cited by the geometry paper but could not be read directly. The Sharpe-based complexity scaling test requires a much larger cross-section ($N \gg 35$) to have statistical power. The P -independence of the spectral gap is consistent with Kelly but is not a direct test of his main theorem.

k_0 is estimated, not formally identified

The spectral gap 98% collapse at $k=8 \rightarrow 12$ and Sharpe peak at $k=8$ both suggest $k_0 \sim 8$ for this 35-stock universe. This is circumstantial evidence. A formal test using Theorem 5.1 — fitting at $k=k_0$ should yield a unique minimiser — has not been implemented.